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Autograde W1M1 question 2c (confidence bounds)

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7 months ago

As I understand the problem, we need to use the principle of error propagation for non-linear functions - which by no means are covered in this course yet :(

(see e.g., https://en.wikipedia.org/wiki/Propagation_of_uncertainty)

My answer is:

```
std_temp = sum(glmmod.apple$coefficients['temp', 'Std. Error'])
```

```
std_d.2.c = sqrt( (diff_quant.2.b * d.test.2.b)^2 * std_temp^2 ); std_d.2.c
```

```
bounds_d.2.c = d.test.2.b + c(-1,1)*qnorm(1-0.05/2)*std_d.2.c; bounds_d.2.c
```

```
temp.odds.lower = bounds_d.2.c[1]
```

```
temp.odds.upper = bounds_d.2.c[2]
```

where

```
d.test.2.b == temp.odds.diff, and
```

```
diff_quant.2.b == temp_1st_quantile - temp_3rd_quantile
```

This should be correct according to the reference, as the derivative of $\exp(\beta \cdot \text{diff})$ is $\text{diff} \cdot \exp(\beta \cdot \text{diff})$, but I do not get credits for the assignment; why???

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